Framing, not structure: an invariance audit of large language model cooperation in the iterated prisoner's dilemma

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Abstract

Large language models increasingly negotiate and coordinate as agents, making their cooperation a governance question. The literature answers with a cooperation rate, or a strategy label like tit-for-tat. Both presuppose that what is measured is a property of the game; we test that presupposition. Across 2,400 iterated prisoner's dilemmas crossing four non-reasoning models, three payoff scales, five languages and four persona pairings, we rescale every payoff by a positive constant - whose irrelevance is a theorem - and restate the rules in another language. Both move cooperation in two of four models, by up to 0.24 and 0.15, while replicates of one prompt disagree more than all manipulations combined. Stranger still, the assigned persona moves behaviour against its own meaning: agents told they are selfish open cooperatively 91% of the time, against 31% told they are cooperative, in every model and language. One hypothesis covers the inversion, the sign of the scale effect and the memory-one fingerprints: the models read the stated penalties as rewards. Strategy labels fare no better; no canonical rule fits 50–86% of agent-games, and widening the vocabulary buys only 0.2 coverage above a shuffled null. Such numbers may describe the prompt rather than the agent.

Keywords: machine behaviour; large language models; iterated prisoner's dilemma; cooperation; framing effects; evolutionary game theory; multi-agent systems

1 Introduction

Large language models (LLMs) are increasingly deployed as *agents*: they negotiate, coordinate and act on behalf of users in recommendation systems, negotiation tools and multi-agent assistants [1, 2]. In these settings behaviour emerges from strategic interaction rather than from isolated model outputs, and it is shaped by training, prompting, role assignment and linguistic framing, with direct implications for safety and governance. The proposal that machines be studied as behaving entities - with the observational tools of ethology and the experimental discipline of the behavioural sciences - has become considerably more urgent as the population of interacting artificial agents has grown [3].

The iterated prisoner's dilemma is the natural testbed for that programme. A two-by-two matrix with $T > R > P > S$ and $2R > T + S$ generates a strategic landscape that half a century of work has mapped in detail: reciprocity, forgiveness, provocability and exploitation are defined precisely and their evolutionary fates computed [4–9]. The same landscape underpins a large body of work on the statistical physics of cooperation [10–12] and on mechanisms such as intention recognition and commitment that displace the cooperative equilibrium [13, 14]. It is no surprise that the map has been imported wholesale. A rapidly expanding literature now evaluates LLMs through game-theoretic lenses, reporting systematic departures from Nash equilibria, persistent cooperative biases and sensitivity to contextual factors such as language and incentives [15–19], alongside probes with the instruments of cognitive psychology [20] and the use of models as simulated economic subjects [21–23].

Read across, that literature reports one of two things. A scalar: model X cooperates at rate p. Or a label: model X plays tit-for-tat. Both are useful summaries, and both carry a presupposition that has not been established. A cooperation rate is a property of an agent only if it is stable under transformations that leave the problem unchanged; otherwise it is a property of the particular way the problem was written down. A strategy label is informative only if the behaviour it names was in fact generated by something in the family from which the label was drawn; otherwise it reports the resolution of the classifier that assigned it. Neither presupposition is safe for a system whose input is a natural-language string and whose behaviour is known to move with features of that string that carry no task content - option order, formatting, an assigned persona [24–27].

This paper takes those presuppositions as its object of study rather than as its foundation, and the instrument is an invariance audit. We adopt two principles as the test. The first is the affine invariance of von Neumann–Morgenstern utility: multiplying every payoff by the same positive constant leaves best replies, equilibria and evolutionary dynamics unchanged [28, 29]. The second is the invariance of a game to the language in which its rules are stated; a prisoner's dilemma described in French is the same prisoner's dilemma. Manipulations built on these principles have a property that is unusual in behavioural work: the null hypothesis is a theorem rather than a convenient default. There is no effect size to argue about and no prior to elicit, and any movement at all is a failure - of the agent, or of the measurement - rather than one more observation to be added to the tally.

The value of that framing is easiest to see against the closest neighbours. Lorè and Heydari [19] separate game structure from contextual framing and find that framing carries comparable weight; Fontana et al. [17] recover rich strategic archetypes from long-horizon play; Akata et al. [15] report stable, model-specific cooperative signatures across repeated games. Each of these results is stated in the vocabulary whose applicability we test - a rate, or an archetype - so what follows is not a competing measurement but a prior question about what such measurements are measurements of.

One regularity in this corpus is easy to state and hard to place: in three of the four models, cooperation rises with payoff magnitude, where an evolutionary game theory (EGT) account of the same matrix predicts the opposite, that defection should take over as stakes grow. The reading such a result invites is that alignment training and the cooperative bias of human training data pull the models away from the EGT baseline in the prosocial direction, and we take that reading seriously in the discussion. The question we put first, however, is prior to it: whether the movement is a strategic response at all. Three questions organise the work:

1. Do LLM agents satisfy the invariances that any game-theoretic description of them would

require, and does the answer differ across models and languages?

2. How much of the variance in cooperation does the experimental design explain, relative to the variance between replicates of an identical prompt?
3. Is the resulting play generated by a strategy in the standard sense, and what mechanism accounts jointly for the invariance failures?

The contributions are: **(i)** an invariance audit of payoff magnitude and prompt language in which the null is exact, including a language test that holds the objective framing fixed and thereby separates a language effect from a framing effect (§3.2–§3.3); **(ii)** the finding that the assigned persona inverts, moving behaviour opposite to its semantic content at the opening move in every model and every language (§3.4); **(iii)** a single account - that the models read the stated penalties as rewards - which explains the persona inversion, the direction of the payoff-scale effect and the shape of the memory-one fingerprints together (§3.5); and **(iv)** a deductive audit of strategy attribution showing that in half to seven-eighths of agent-games none of the four canonical rules fits at all, so that an archetype label reports the resolution of the classifier rather than the behaviour of the model, and that widening the vocabulary does not rescue it - all 32 deterministic memory-one rules plus one regime switch still leave 15–42% unexplained and exceed a shuffled null by only about 0.2, because the play conditions on two rounds of history rather than one (§3.6–§3.7).

2 Methods

The design is built backwards from the tests. Each factor we cross - payoff magnitude, prompt language, assigned persona - is one whose effect on an agent that is playing the game is known before any data are collected, so the corpus is sized and balanced to detect movement precisely where the prediction is no movement at all. Everything the manipulations do not touch is held fixed: the same stage game, the same horizon, the same interface between agent and game, the same decoding settings.

2.1 The game and its manipulations

FAIRGAME (Framework for AI Agents Bias Recognition using Game Theory) [30] provides the computational infrastructure. Experimental conditions are defined via JSON configuration files specifying payoff structures, game horizon, LLM backends and languages; at runtime FAIRGAME combines these with language-specific prompt templates, simulates repeated normal-form games and logs round-by-round trajectories. We adopt it for two reasons beyond convenience: its multilingual templates are the object of the language manipulation rather than an incidental choice of wording, and taking them as given rather than authoring our own keeps the wording of the manipulation out of our hands. We add payoff scaling for the prisoner's dilemma.

Agents played a symmetric two-player prisoner's dilemma (figure 1a). The templates state the cell values as *penalties* - years of imprisonment in the prisoner's-dilemma narrative - and instruct the agent to *minimise* them, so a smaller number is a better outcome. Under that framing the base cells are $T = 0$, $R = 2$, $P = 6$, $S = 10$, satisfying $T < R < P < S$, which is the penalty-space image of the usual $T > R > P > S$. Option A is therefore the dominant, defecting action and Option B is cooperation: mutual cooperation carries the lowest symmetric penalty $(2, 2)$, mutual defection $(6, 6)$, and defecting on a cooperator carries none at all. On the utility scale $u = -$penalty the two normalised descriptors of the dilemma are greed $(T - R)/(T - S) = 0.20$ and fear $(P - S)/(T - S) = 0.40$.

The penalty framing is inherited from FAIRGAME rather than chosen here, and we retain the templates unchanged rather than rewriting them; §3.5 argues that the choice turns out to be considerably more consequential than it looks. We flag the polarity explicitly because it is easy to invert and because inverting it maps every cooperation rate $x \mapsto 1 - x$ and reverses the sign of both the payoff-scale and the endgame effects. It is not, however, a matter of interpretation. FAIRGAME logs the realised payoff alongside every action, and those values fix the assignment: mutual Option B returns the lower symmetric penalty 2λ to both agents, mutual Option A returns 6λ, and Option A against Option B returns nothing to the defector and 10λ to the victim. This assignment reproduces the logged payoff exactly on all 24,000 rounds of the corpus; the inverted assignment reproduces none of them.

Three factors are manipulated on top of that fixed game, and what distinguishes the first two is that their predicted effect is not small but exactly zero.

Payoff magnitude. Every cell was multiplied by a payoff scale $\lambda \in \{0.1, 1, 10\}$. Because $\lambda > 0$ this is a positive affine rescaling of a von Neumann–Morgenstern utility [28, 29]: the outcome ordering, the best-reply correspondence, the equilibrium set and the replicator dynamics are all unchanged, so the three games are one game presented with different numbers. The realised scale was recovered per game from the payoff values actually recorded rather than assumed from the directory: 80 rounds (0.17% of the corpus) carry $\times 1$ payoffs inside a folder labelled $\times 10$, and are labelled by their realised scale throughout.

Prompt language. The same game was instantiated in English, French, Vietnamese, Chinese and Arabic from the FAIRGAME templates. A game is not changed by the language its rules are stated in, so here too the predicted effect is exactly zero. One caveat is a finding rather than a nuisance, and we treat it as such: the templates do not state the objective identically in every language - English, French and Vietnamese instruct the agent to minimise a penalty, Chinese and Arabic to maximise a reward - so a five-language comparison mixes a language manipulation with an objective-framing manipulation, and every language test is therefore repeated within the three languages that share a single framing (§3.3).

Assigned persona. Each agent received a system-prompt persona, either *cooperative* or *selfish*, and the four ordered pairings were crossed with everything else. This manipulation differs in kind from the other two: a persona is not irrelevant by theorem, and could in principle be read as an instruction about how to play. But it carries no information about the game - the four pairings present identical payoff matrices - and an agent was never told its opponent's persona, so the opponent's persona is a manipulation the focal agent can learn about only through play. Here it is the direction of the effect, rather than its size, that the semantics of the words predict.

2.2 Design and procedure

The design crossed four models - Claude 3.5 Haiku, Gemini 3.5 Flash-Lite, GPT-4o and Mistral Large - with three payoff scales, five prompt languages, four persona pairings and ten replicates: $4 \times 3 \times 5 \times 4 \times 10 = 2{,}400$ dyads, 600 per model (figure 1b). Each dyad ran for ten rounds and both agents are recorded, so the corpus contains 4,800 agent-games and 48,000 binary decisions, with exactly 40 dyads in every model $\times$ scale $\times$ language cell. The cell is the unit of the design: none is more replicated than another, and no result below depends on pooling across an unbalanced margin.

On each round an agent received the full payoff table, its persona, the round index and the complete history of both players' choices, and returned a single token naming one of two options. Communication was disabled, so the only channel between agents is the action history. Actions were labelled `OptionA` and `OptionB` in all languages, avoiding the connotations that the words "cooperate" and "defect" would carry; which label denotes cooperation was never stated to the agent, and has to be read off the payoff table.

Decoding used a fixed low output-token budget with reasoning traces disabled. This is not cosmetic: models that emit hidden reasoning can exhaust a small output budget inside the trace and return a truncated string that fails to parse, and a harness that then falls back to a default action produces an apparent cooperation rate of 1.0 that is an artefact of the parser rather than a measurement of the model. Parse-failure and fallback rates were logged for every cell and required to be below 0.02. Sampling temperature follows each provider's recommended default, as in the FAIRGAME protocol - 1.0 for Claude, Gemini and GPT-4o, 0.3 for Mistral Large.

Two features of the design are confounded with model identity, and we declare them here rather than leave them to be discovered. The Gemini arm was run with the round count disclosed to the agents; the Claude, GPT-4o and Mistral arms with the horizon hidden, so the Gemini column is a "model $\times$ horizon" column and is marked as such in figure 1b. Sampling temperature, fixed by the provider default above, likewise cannot be separated from model identity; because it is constant within a model it cannot generate any of the within-model effects reported below, but between-model comparisons carry it. A third candidate, the fact that the templates do not state the objective identically in every language, is a result rather than a nuisance and is treated in §3.3.

2.3 Measures and strategy read-out

The primary measure is the cooperation rate, the proportion of a game's ten rounds on which an agent chose Option B. Because both agents of every dyad appear as rows, the exploited (CD) and exploiting (DC) shares are equal by construction and are merged into a single anti-coordination band in composition analyses.

The memory-one fingerprint is the vector $(p(C \mid R), p(C \mid S), p(C \mid T), p(C \mid P))$ of probabilities of cooperating after each joint outcome of the previous round. This is the complete description of a memory-one strategy and places every canonical rule at a known point: AllC at $(1,1,1,1)$, AllD at $(0,0,0,0)$, tit-for-tat at $(1,0,1,0)$, win-stay lose-shift at $(1,0,0,1)$ and grim trigger at $(1,0,0,0)$. Niceness is $p(C)$ on round one and reciprocity is $p(C \mid C_{-1}) - p(C \mid D_{-1})$.

Assigning a strategy label was done in two stages, kept distinct because they license different claims. The first is exact and deductive: for each agent-game we asked which of the four canonical rules - AllC, AllD, tit-for-tat and win-stay lose-shift - reproduces every observed transition. A game may be matched by exactly one, by several, or by none, and only the first case supports the statement that the agent played that rule. Matching by several is an answer rather than a failure: an agent that cooperates for ten rounds against a cooperator *is* AllC and tit-for-tat and win-stay lose-shift at once, so we report the set and do not break the tie. The second stage is inductive and applies only where the first returns nothing: a long short-term memory classifier [31] trained on synthetic play from known generating strategies returns a posterior over archetypes, whose argmax we take as the nearest neighbour. We also record, for each agent-game, the minimum number of rounds on which play departs from its own nearest rule. Because the two stages license different claims, we never report a single pooled label distribution: the share reached deductively and the share reached by nearest neighbour are always separated (figure 5c).

The four canonical rules are a small vocabulary, and a rule may fail to fit simply because the vocabulary is too narrow rather than because the play has no rule in it. We therefore widen the hypothesis class in controlled steps and pair every coverage figure with a null (§3.7): first to all 32 deterministic memory-one rules, then to trajectories explained by two such rules in sequence with a single switch between them. The null shuffles the focal agent's own action sequence in place, holding its cooperation rate and the opponent's realised sequence fixed. It is not a formality: a two-segment fit over ten rounds has enough freedom to reproduce a large share of shuffled play by chance, so raw coverage on its own is not evidence of structure. Coverage is always reported against the vocabulary that produced it and the two are never pooled - the share of games matched by the canonical four and the share matched by the 32-rule class are different quantities, and they differ by an order of magnitude once ambiguity is taken into account.

A vocabulary can fail in two ways: by being too small, or by conditioning on the wrong thing. To separate them we ask how much history the next action actually uses, scoring nested predictors of the next action - memoryless, memory-one, memory-two, and a memory-one rule allowed to change with the phase of the game - against one another by the Bayesian information criterion [32] on identical rows, so that the comparison is about history rather than sample size.

2.4 Statistical analysis

All intervals are 95% bootstrap confidence intervals over 2,000 resamples that resample whole dyads [33]. This is consequential: rounds within a game are autocorrelated and the two agent-rows of a dyad share a history, so a row-level resample would understate the interval by roughly $\sqrt{2}$. The payoff-scale slope is the change in cooperation per decade of λ, fitted on $\log_{10} \lambda$. Language disparity is the difference between a model's highest- and lowest-cooperation language, tested against a permutation null that shuffles the language label across dyads within a model over 5,000 draws. The variance decomposition is sequential (Type I) in the order model, payoff scale, language, persona; entering model identity first makes its share an upper bound.

Three reliability checks were run whose expected answer is "nothing here". Agents 1 and 2 of a matched pairing receive identical prompts and their cooperation rates correlate at $r = 0.67$–0.95 with a mean position bias of $+0.017$. The ten replicates of a cell share a prompt and their spread is 1.3–1.6 times what independent rounds would produce. Cell means reproduce across a random split of the dyads with Spearman–Brown reliability of 0.90–0.97.

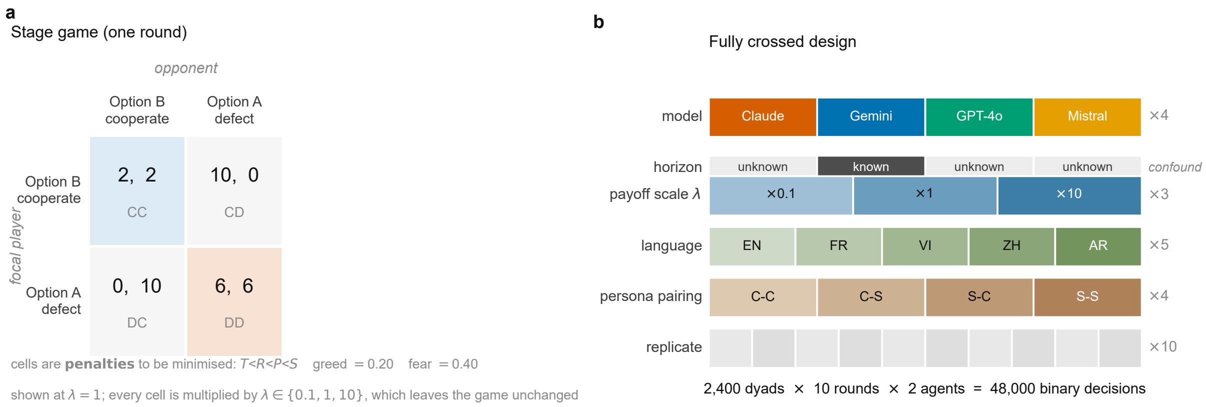

Figure 1: The game and the design. (**a**) The stage game at $\lambda = 1$. Cells are penalties the agent is told to minimise, so $T < R < P < S$ and Option A is the dominant, defecting action. Greed and fear are computed on the utility scale $u = -\text{penalty}$. Every cell is multiplied by $\lambda \in \{0.1, 1, 10\}$, a positive affine rescaling that leaves the game strategically identical. (**b**) The fully crossed design: 2,400 dyads, 48,000 binary decisions, with 40 dyads in every model × scale × language cell. The horizon row is shaded as a confound rather than a factor, because only the Gemini arm was told the round count.

3 Results

3.1 Aggregate cooperation conceals more than it reports

Pooled across the design, the four models cooperate at very different rates: 0.74 for Gemini 3.5 Flash-Lite, 0.63 for Claude 3.5 Haiku, 0.52 for Mistral Large and 0.45 for GPT-4o. Read naively this is the headline of a conventional benchmark report, and it is the form in which such results are usually communicated.

It is worth establishing immediately how little that number carries. The distribution of within-game cooperation is not concentrated near its mean for any model (figure 2a). Mistral Large plays 25.0% of its games entirely cooperatively and 24.3% entirely defecting, so its mean of 0.52 describes a bimodal mixture rather than a typical game; Gemini reaches 36.9% wholly cooperative games and GPT-4o 14.9% wholly defecting ones.

How much of this spread does the design account for? Very little (figure 2b). In a sequential decomposition, model identity explains 9.5% of the variance in dyad-level cooperation, persona pairing 5.5%, payoff scale 3.6% and language 0.2%, for a total of 18.7%. The remaining 81.3% is within-cell: ten replicates of a byte-identical prompt disagree with one another more than all four deliberate manipulations together move the mean. Because model identity was entered first, its 9.5% is an upper bound. This is the context in which the invariance tests should be read: we are not asking whether the models differ, but whether the differences that manipulations produce are differences a strategic agent could have. Table 1 summarises the per-model profile developed below.

3.2 Payoff magnitude changes behaviour

Multiplying every payoff by a positive constant leaves best replies, equilibria and evolutionary dynamics unchanged, so the null for a strategic agent is not approximately zero but exactly zero.

Two of the four models violate it decisively (figure 3a). Claude 3.5 Haiku gains 0.120 cooperation per decade of λ (95% CI [0.102, 0.137]) and GPT-4o gains 0.117 ([0.094, 0.140]),

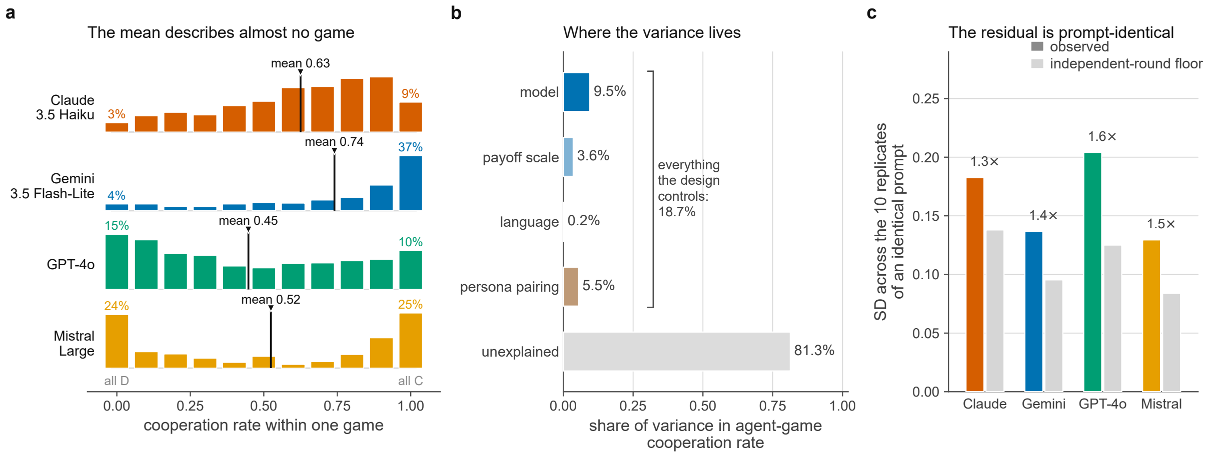

Figure 2: What the pooled cooperation rate hides. (**a**) The distribution of within-game cooperation. Ten rounds admit only eleven values, so this is the discrete distribution, not a histogram. Each row is normalised to its own maximum, so shapes are comparable but heights are not counts. The vertical rule marks the pooled mean, drawn through the distribution it is meant to summarise; the coloured percentages give the share of games played entirely one way. (**b**) Sequential (Type I) variance decomposition of the agent-game cooperation rate, entered in the order shown, so each factor is credited only with what the factors above it leave unexplained. Entering model identity first makes its 9.5% an upper bound. (**c**) What the unexplained share is made of. Bars give the standard deviation of cooperation across the ten replicates of a byte-identical prompt, against the floor that ten independent rounds would already impose; the annotation is their ratio.

total swings of 0.24 and 0.23. Gemini shows a smaller but non-zero slope of 0.030 ([0.007, 0.053]), and only Mistral Large is consistent with invariance at 0.015 ([−0.012, 0.043]).

The direction is the one flagged in the introduction: cooperation *rises* with the stated stakes, where an evolutionary account of the same matrix predicts that defection should take over as stakes grow. What the invariance framing adds is that the movement cannot be a strategic response in either direction, because there is no strategic difference to respond to. Note also that the effect is not monotone in the way a magnitude heuristic would predict: for Claude and GPT-4o almost the entire swing occurs between ×0.1 and ×1, with ×1 and ×10 nearly indistinguishable, which points to sensitivity to the appearance of the numbers - fractional versus integral, single- versus multi-digit - rather than to magnitude as such.

3.3 Prompt language changes behaviour, with framing held fixed

Measured across all five languages, three of four models show a language disparity beyond its permutation null: 0.156 for Gemini ($P < 0.001$), 0.155 for Mistral ($P < 0.001$) and 0.087 for Claude ($P = 0.001$), against 0.040 for GPT-4o ($P = 0.70$).

That comparison is confounded, and we treat the confound as a result. The English, French and Vietnamese templates instruct the agent to minimise a penalty, whereas the Arabic and Chinese templates instruct it to maximise a reward while still calling the outcomes penalties. A raw five-language comparison therefore mixes a language manipulation with an objective-framing manipulation, and the language effect must be re-tested with framing fixed.

Doing so within English, French and Vietnamese alone (figure 3b, solid bars) leaves the effect intact for two of four models: Mistral Large retains a disparity of 0.152 ($P < 0.001$) and Claude 0.087 ($P = 0.002$), both above their permutation nulls. Gemini's apparent disparity

Table 1: Behavioural profile of each model. Cooperation is pooled over the whole design. The payoff-scale slope is the change in cooperation per decade of λ with 95% bootstrap CI; its null is exactly zero. The language gap is measured within the three languages sharing one objective framing. "Opening C" gives p(cooperate on round 1) for an agent told it is cooperative and for one told it is selfish; the ordering is reversed in every model. Distance is Euclidean, in the four-dimensional memory-one fingerprint space, to the nearest canonical rule, which are themselves at most 2.0 apart. "Exact rule" is the share of agent-games in which exactly one of the four canonical rules reproduces the whole trajectory; the wider 32-rule vocabulary is treated in §3.7.

Model	Coop. rate	Scale slope (per decade)	Lang. gap (fixed frame)	Opening C coop.	Opening C selfish	Nearest rule (dist.)	Exact rule
Claude 3.5 Haiku	0.63	+0.120 [0.102, 0.137]	0.087**	0.14	0.81	TFT (0.76)	9.7%
Gemini 3.5 Flash-Lite	0.74	+0.030 [0.007, 0.053]	0.052	0.54	0.98	GRIM (0.90)	21.9%
GPT-4o	0.45	+0.117 [0.094, 0.140]	0.028	0.38	0.84	GRIM (0.71)	21.8%
Mistral Large	0.52	+0.015 [−0.012, 0.043]	0.152***	0.19	1.00	GRIM (1.01)	48.7%

$^{**}P < 0.01$, $^{***}P < 0.001$ against a permutation null; unmarked gaps do not exceed their null.

collapses from 0.156 to 0.052 and ceases to be significant ($P = 0.19$), identifying most of its five-language gap as a framing effect rather than a language effect. GPT-4o remains invariant on both readings ($P = 0.64$). The framing contrast itself is small and inconsistent in sign across models (−0.093 to +0.064), so it does not act as a simple main effect either.

The surviving effects are substantial: Mistral cooperates at 0.61 in English and 0.46 in Vietnamese - three languages, one objective framing, one payoff matrix, a gap of 0.15. We emphasise that this is a claim about two of four models rather than three, and that the more cautious claim is the one the data support. Because the disparity is a range, it is invariant under $x \mapsto 1 - x$ and so is the one result in this paper that does not depend on the polarity convention of §2.1 at all.

3.4 The assigned persona inverts

The persona manipulation differs from the previous two in that it is not irrelevant by theorem; a persona could in principle be read as information. But an agent's own persona carries no information about the game, and the four pairings present identical payoff matrices. What we find is not that the persona fails to move behaviour, but that it moves it the wrong way.

Pooled over the corpus, an agent told it is cooperative cooperates 0.517 of the time and an agent told it is selfish 0.651. The effect is concentrated in the opening move and is then eroded by the interaction: at round one the two groups cooperate at 0.312 and 0.908 respectively, a gap of −0.60; by round ten the gap has closed to +0.01. The inversion at the opening holds in every model - Claude 0.14 against 0.81, Mistral 0.19 against 1.00, GPT-4o 0.38 against 0.84, Gemini 0.54 against 0.98 - in every language, with gaps from −0.42 to −0.73, and at every payoff scale, with gaps of −0.81, −0.49 and −0.48 at $\lambda = 0.1, 1, 10$. Mistral Large told it is selfish opens cooperatively in 100% of its 600 games; told it is cooperative, in 19%.

At the level of the whole game the pooled effect remains negative for three of four models (figure 3c): −0.354 for Mistral ([−0.408, −0.303], $d = -0.94$), −0.158 for Claude ([−0.189, −0.127], $d = -0.61$) and −0.045 for Gemini ([−0.086, −0.001]), with GPT-4o indistinguishable from zero (+0.018, [−0.028, 0.067]).

One further observation complicates any reading in which persona simply installs a disposition.

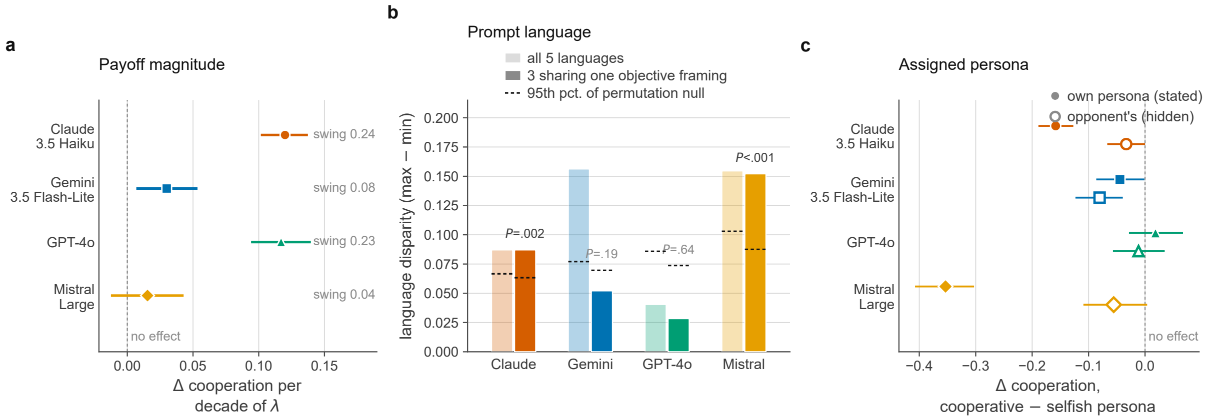

Figure 3: Three manipulations that cannot matter, and all three do. (**a**) Change in cooperation per decade of the payoff scale λ, with 95% bootstrap confidence intervals resampling whole dyads. A positive affine rescaling leaves the game strategically identical, so the null is exactly zero. Claude and GPT-4o depart from it by more than 0.11 per decade; the annotation gives the total swing across the three scales. (**b**) Language disparity, the gap between a model's highest- and lowest-cooperation language. Pale bars use all five languages, in which objective framing varies with language; solid bars use only English, French and Vietnamese, which share a single framing. Dashes mark the 95th percentile of a permutation null; P values are for the framing-controlled comparison. (**c**) Effect of the assigned persona. Filled markers give the difference between an agent told it is cooperative and one told it is selfish; open markers give the same difference for the *opponent's* persona, which is never disclosed. Three of four filled markers lie left of zero: the instruction moves behaviour opposite to its semantic content.

The opponent's persona - never disclosed, inferable only from play - has a detectable effect for Gemini, which cooperates 0.080 less against a cooperative opponent, and marginally for Claude. For Gemini this indirect effect is larger than its own-persona effect, which is the signature of an agent responding to the interaction rather than to the instruction.

3.5 One account of all three: the frame is misread

The three failures above look like three separate sensitivities. They are jointly explained by a single hypothesis: that the models treat the displayed numbers as rewards to be maximised rather than as penalties to be minimised, pattern-matching the matrix to the canonical reward-framed prisoner's dilemma in which the largest entry is the prize.

Under a reward reading of the same table (figure 4a), Option A yields the symmetric pair $(6, 6)$ and Option B can yield the single largest number, 10. An agent told it is cooperative and reaching for the outcome that looks mutually good would therefore choose A; an agent told it is selfish and reaching for the largest attainable number would choose B. Both are exactly what we observe (figure 4b), and both are the opposite of what the stated objective implies, because under minimisation A is the dominant defecting action and the 10 is the sucker's penalty. The hypothesis also predicts the sign of the payoff-scale effect: enlarging λ magnifies the numbers and makes the salient maximum more conspicuous, so the share of B should rise with λ, which is what figure 3a shows.

The memory-one fingerprints carry the same signature (figure 4c). All four models cooperate readily after having been exploited - $p(C \mid S)$ is 0.48, 0.81, 0.63 and 0.96 for Claude, Gemini, GPT-4o and Mistral - where tit-for-tat, grim and win-stay lose-shift all prescribe 0. Being

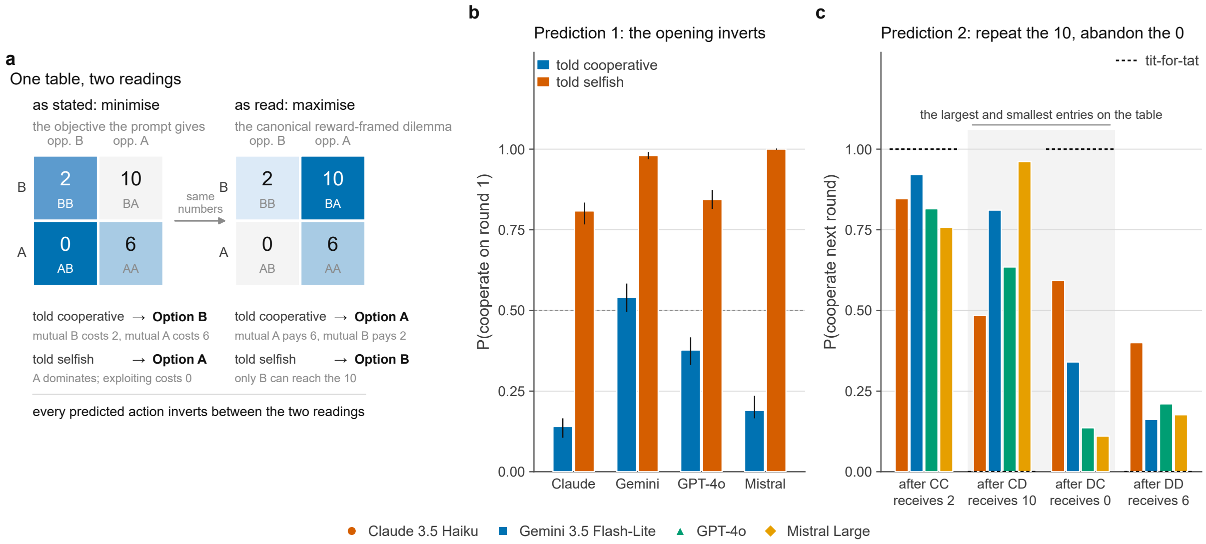

Figure 4: One account of all three failures: the frame is read backwards. (**a**) The same four numbers under the two readings, shaded by how good each cell is for the focal player, whose action indexes the rows. Under the stated objective the agent minimises a penalty, so the 0 is the prize; under a reward reading the 10 is. Every action the two readings prescribe is the inverse of the other, for both personas. (**b**) The prediction at the opening move, where no history exists and the prompt is the only input. Agents told they are selfish open cooperatively far more often than agents told they are cooperative, in all four models, with 95% bootstrap intervals resampling whole dyads. (**c**) The prediction inside the memory-one fingerprint. State labels carry the penalty the focal agent received. All four cooperate readily after CD, where the agent has just collected the largest entry on the table and every canonical rule except AllC prescribes 0; three of four cooperate less after DC, where it collected nothing. Dashes mark tit-for-tat.

"exploited" in this corpus means receiving the penalty 10, the largest number on the table; an agent that has just been handed the biggest number and repeats the action that produced it is behaving consistently under a reward reading and inexplicably under the stated one. Symmetrically, three of four cooperate *less* after having successfully exploited the opponent, where the payoff was 0: $p(C \mid T)$ is 0.34, 0.14 and 0.11 for Gemini, GPT-4o and Mistral against tit-for-tat's 1. Mistral's reciprocity index is accordingly negative, -0.149.

We present this as the most parsimonious account rather than as a demonstration. A competing explanation is ordinal: "Option A" is listed first, and a cooperative persona may be reaching for the default rather than for the larger symmetric number. Distinguishing the two requires a manipulation we did not run - re-labelling or reordering the options, or restating the identical matrix in explicit reward form and testing whether behaviour mirrors - and we regard that as the natural next experiment. What the present data do establish is that the stated objective is not what the models are optimising.

3.6 The play is not a strategy

Endgame. With a finite horizon, backward induction predicts decay towards mutual defection. Three of four models show it (figure 5a): GPT-4o falls from 0.61 to 0.41 between rounds one and ten (-0.20), Gemini from 0.76 to 0.68 (-0.08) and Mistral from 0.60 to 0.52 (-0.08). Claude is the exception, rising from 0.47 to 0.74 ($+0.27$). The exception is not a marginal one: Claude's rise exceeds any of the three decays in magnitude, so a panel-level endgame trend

Figure 5: What the play is not. (**a**) Cooperation by round with 95% bootstrap bands. Three of four models decay towards the endgame, consistent with backward induction; Claude is the exception, rising from 0.47 to 0.74. Gemini is the single arm that was told the round count. (**b**) Euclidean distance from each model's memory-one fingerprint to every canonical rule, not only to the nearest one, since "nearest" hides whether the runner-up was a close second. Markers ring each model's nearest rule. The dashed line is the greatest distance between any two canonical rules in this space, which is the scale on which 0.7–1.0 should be read. (**c**) How an archetype label is actually reached. Dark: exactly one of the four canonical rules reproduces the whole trajectory. Mid: several do, and the game does not distinguish them. Grey: none does, and the label is a nearest-neighbour statement from the classifier. The right-hand column gives the mean number of rounds out of ten on which play departs from its own nearest rule.

would average to almost nothing and would report the composition of the panel rather than a shared mechanism.

No memory-one rule. The fingerprints place every model far from every canonical strategy (figure 5b). The nearest archetype is tit-for-tat for Claude at Euclidean distance 0.76, and grim trigger for GPT-4o (0.71), Gemini (0.90) and Mistral (1.01). For scale, the canonical rules are at most 2.0 apart from one another in this space, so a distance of 0.7–1.0 is not a near miss. Nor is the nearest rule nearest by much: the runner-up is only 0.11 to 0.33 further away, a margin smaller in every model than the distance to the nearest rule itself, so which archetype a label names is decided by much less than the error that label already carries.

A label is mostly not a description. Exactly one of the four canonical rules reproduces the entire trajectory in 9.7% of Claude's agent-games, 21.8% of GPT-4o's, 21.9% of Gemini's and 48.7% of Mistral's (figure 5c). A further 4.7%, 5.8%, 21.7% and 1.2% are matched by more than one rule at once, which the ten rounds do not separate. In the remaining 85.7%, 72.4%, 56.4% and 50.2% no canonical rule fits and the label is a nearest-neighbour statement; where a label is assigned this way, play departs from the rule it names on 1.0 to 2.0 rounds out of ten.

3.7 What the unmatched play is

The result above is negative, and a negative result of this kind invites an easy dismissal: that the residual is noise, and no label was ever going to fit noise. It is worth answering, because the answer decides whether the strategy vocabulary is too small or simply the wrong one.

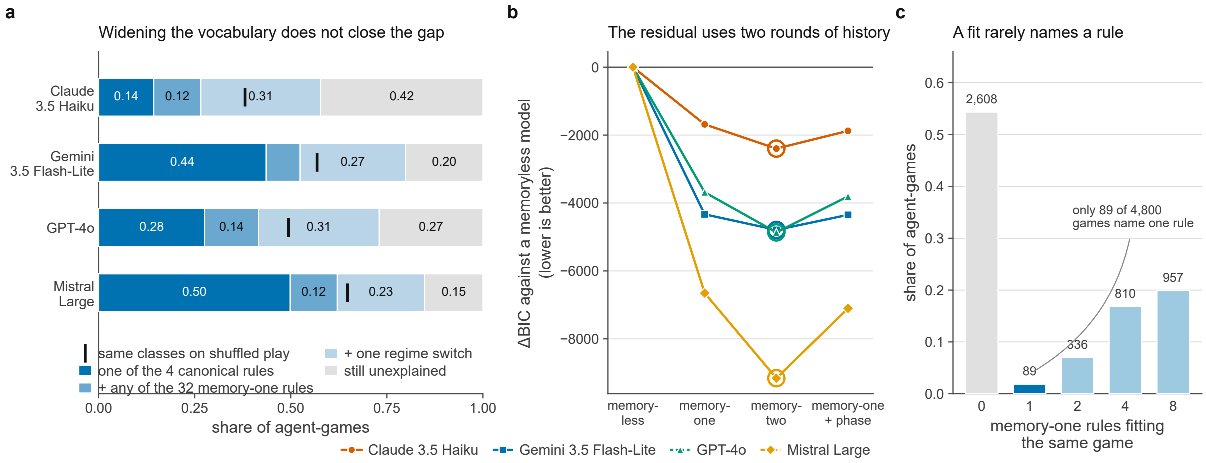

Figure 6: The residual is real, and it is not in the class. (**a**) Coverage as the hypothesis class is widened. The classes are nested, so the segments are disjoint and stack to one. The tick is where the same three classes reach on shuffled play, in which the focal agent's own action sequence is permuted in place while its cooperation rate and the opponent's realised sequence are held fixed; coverage to the left of the tick is what the fit buys by chance. (**b**) How much history the next move uses. ΔBIC of nested predictors of the next action against a memoryless model, scored on identical rows so the comparison is about history and not sample size. The ring marks each model's selected depth; all four select memory-two. (**c**) Why even a fit rarely names a rule. Ten rounds seldom visit all four conditioning states, so rules differing only in an unvisited state are indistinguishable on that game. Bars pool all 4,800 agent-games.

Widening the class does not close the gap (figure 6a). Admitting all 32 deterministic memory-one rules rather than the canonical four adds 12.3, 8.8, 14.0 and 12.3 percentage points for Claude, Gemini, GPT-4o and Mistral; admitting trajectories built from two such rules with a single switch between them adds a further 31.2, 27.4, 31.4 and 22.8. Even then 42.3% of Claude's agent-games, 27.0% of GPT-4o's, 20.2% of Gemini's and 15.2% of Mistral's are not reproduced by anything in the class. And the coverage that is achieved must be read against its null: the same three classes fit shuffled play up to 0.38, 0.57, 0.49 and 0.65, so the excess attributable to structure rather than to the flexibility of a two-segment fit over ten rounds is only 0.20, 0.23, 0.24 and 0.20. Roughly one agent-game in five is explained by the widened vocabulary beyond chance.

The reason is not that the play is unstructured but that the structure uses more history than the class allows (figure 6b). Scoring nested predictors of the next action on identical rows, all four models select memory-two: ΔBIC against a memoryless baseline improves from −1685 to −2399 for Claude, −4334 to −4792 for Gemini, −3684 to −4865 for GPT-4o and −6652 to −9158 for Mistral. Allowing the memory-one rule to change with the phase of the game instead - an equally sized extension in the direction of time rather than of history - is worse than memory-two for every model. Whatever these agents are doing conditions on the last two rounds, so a memory-one vocabulary is not a coarse description of it but a description of something else.

A final limit applies to the deductive fraction itself (figure 6c). Over ten rounds a trajectory rarely visits all four conditioning states, so rules that differ only in an unvisited state cannot be told apart. Of the 4,800 agent-games, 2,608 match no memory-one rule, and of those that do match, most are matched by four or eight rules at once; exactly 89 - 1.9% of the corpus - are pinned to a single one of the 32. A coverage figure for the wider class is therefore a statement about a *set* of rules and not about a rule, and the ten-round horizon is what makes it so.

3.8 What is stable

It would be a misreading to conclude that nothing here is measurable. The quantities we compare are reliable even though individual games are not: split-half reliability of the cell means is 0.90 for Claude, 0.91 for GPT-4o, 0.95 for Gemini and 0.97 for Mistral. What is unstable is the individual game. Ten replicates of an identical prompt have a standard deviation 1.3 to 1.6 times what independent rounds would produce (figure 2c). The correct summary is that these systems have reproducible *propensities* and irreproducible *episodes*.

4 Discussion

4.1 What an invariance failure means

The central results are failures of two invariances that are not empirical regularities but consequences of how games are defined. Two of four models move by more than 0.11 in cooperation per decade of payoff scale, and two of four move by 0.09–0.15 across languages sharing an objective framing. The consequence is not that the models are irrational in the ordinary sense of choosing dominated actions; it is more specific. The object that a cooperation rate is supposed to measure is not well defined for these systems, because the measurement changes when the thing measured does not. Reporting that a model cooperates at 0.45 without reporting that the same model cooperates at 0.29 or 0.53 depending on whether the payoffs are written as decimals or as hundreds is not a partial description but a description of an artefact of presentation.

This connects the machine-behaviour programme [3] directly to the framing literature in human decision making [34, 35]. The disanalogy is instructive. Human framing effects are systematic, map to a compact set of psychological mechanisms and reproduce across populations. What we observe is heterogeneous across models - Mistral is scale-invariant but the most persona-inverted; GPT-4o is language-invariant and persona-inert but strongly scale-sensitive - and sits on top of a within-cell variance that dwarfs it. There is no single “LLM framing effect” to characterise, only a per-model, per-manipulation profile that must be measured.

4.2 Reinterpreting the cooperative bias

The natural reading of the rise of cooperation with stakes, and the one we began with, is that it is evidence of alignment training and of the cooperative bias of human training data, the LLM distribution pulling away from an EGT baseline in the prosocial direction. The persona inversion suggests a complementary and less flattering mechanism. If the models are reading the penalties as rewards, then part of what has been measured as prosociality is number-chasing: the action that registers as “cooperation” under the correct framing is also the action carrying the largest single entry on the table, and it becomes more attractive precisely as λ makes that entry larger.

We do not claim this displaces the alignment account. The two are separable by experiment - restate the identical game in explicit reward form, in which the cooperative action no longer carries the largest number, and see whether the stake effect survives - and we regard that as the most valuable single extension of this work. What the present data do establish is that a cooperation rate measured under a penalty framing cannot be read as prosociality without that control.

4.3 Consequences for strategy attribution and for hybrid populations

A large part of the appeal of the prisoner's dilemma is that its strategy space is well mapped, and it is tempting to import that map wholesale. Our results suggest the import is not licensed. A statement of the form "model X plays tit-for-tat" should be accompanied by the fraction of agent-games in which a rule was deductively established rather than assigned by nearest neighbour, and by the number of rounds on which play departs from its assigned label. On our corpus those numbers are 10–49% and 1.0–2.0 rounds out of ten.

The natural repair - enlarge the strategy set until the labels fit - does not work either, and the way it fails is informative. Going from four rules to all 32 memory-one rules, and then to two of them in sequence, leaves 15–42% of agent-games unexplained and buys only about 0.2 of coverage above a shuffled null, while the selected memory depth is two rounds rather than one for every model. The class is not too small; it is the wrong class. This matters for the hybrid read-out that is becoming standard practice in this literature, in which a learned classifier supplies a label wherever exact matching fails. Such a classifier will always return something, and on our corpus it is returning something for the majority of games - 50% to 86% depending on the model. The remedy is not a better classifier but reporting the provenance of each label, which costs nothing once the deductive stage is run.

The results also bear on efforts to model populations in which artificial and human agents interact [2, 36, 37]. Such models are typically built by assigning agents strategies from a known family and computing the resulting dynamics. If an artificial agent's cooperation depends on the units in which payoffs are written, its behaviour is not a function of the game and cannot be carried across a replicator equation that only knows the game: two population models identical up to a rescaling would make different predictions about the same agent. And if 81% of the variance is within-cell, the appropriate representation is not a strategy but a distribution over strategies, possibly one whose support changes with framing.

4.4 Limitations

Four limitations bound these conclusions. The model set is the most consequential: four models is a small sample of a fast-moving population, and nothing here establishes that the pattern generalises to models with explicit reasoning procedures, which may well perform the minimisation the present models appear to skip. Testing the invariances on reasoning models is the single most valuable extension, and we regard these results as establishing the diagnostic rather than settling the question.

The frame-misreading account is an inference from converging evidence, not a controlled demonstration; the option-ordering alternative in §3.5 remains open until the relabelling experiment is run.

The horizon confound remains: the Gemini arm knew the round count and the others did not, so that column mixes model identity with horizon knowledge. And the framing-controlled language analysis rests on three languages rather than five, which reduces power, so the two null results should be read with that in mind. Finally, ten rounds is short, and this bounds §3.7 in particular: it under-identifies the memory-one vocabulary, so most matched games are matched by a set of rules rather than by one; it gives a two-segment fit enough freedom that a shuffled null already reaches 0.38–0.65, which is why every coverage figure is reported against that null; and it compresses any endgame effect into a small number of rounds. The memory-two selection is the one result here that a longer horizon would sharpen rather than merely confirm.

4.5 Conclusion

Four frontier language models playing 2,400 iterated prisoner's dilemmas fail invariances that any game-theoretic description of them would require. Rescaling payoffs, which cannot change the game, moves cooperation by up to 0.24. Changing the prompt language with the objective framing held fixed moves it by up to 0.15. And the assigned persona moves behaviour opposite to its semantic content: told they are cooperative, agents open cooperatively 31% of the time; told they are selfish, 91%. A single account - that the models read the stated penalties as rewards - covers the inversion, the direction of the scale effect and the shape of the memory-one fingerprints.

We therefore propose a minimal standard. A cooperation rate reported for a language model should be accompanied by its sensitivity to payoff scale and to prompt language; a strategy label should be accompanied by the fraction of games in which that label was deductively established; and any study framing its payoffs as penalties should verify that the models are optimising the stated objective before reading the result as prosociality. These are cheap to compute and they change what the numbers mean. Until they are routine, the field risks building an evolutionary game theory of artificial agents on quantities that are properties of prompts rather than of agents.

Data accessibility

The full corpus of 2,400 dyads, the analysis pipeline that produces every figure and table, and the supplementary figure suite are available at [REPOSITORY URL]. Every number appearing in a figure is also written to a machine-readable CSV file in the same repository. A permanent archive with a DOI will be deposited on acceptance.

Competing interests

The authors declare no competing interests.

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